Saathi: a calm paper-and-medicine companion for older adults

Working title · Project blueprint for Neighborhood Hacks · Prepared 20 Oct 2026 for Naman · Version 1 (research and design, before building)

1. Summary

Elevator pitch. Saathi lets an older person photograph a medicine strip, a prescription or an official paper, then explains it out loud in their own language, calmly. It first asks how they are and what they already know, explains in small steps, and warns when a paper looks like a scam.

Tracks it fits. Accessibility and Inclusion (primary), Health and Well-being, Safety and Emergency Response (scam protection), Civic Engagement and Education and Access (understanding official paper).

What makes it different from a "scan a medicine" app.

- It treats fear as a design problem: the explanation is a conversation, not a data dump.
- One companion for the two kinds of paper older people struggle with most: medicines and official notices.
- Facts come from verified data; the small AI model only talks, asks and rewrites.
- It catches duplicate ingredients across strips and checks notices for scam patterns.

Core loop. Photo → read and confirm → ask gently → explain in layers → read aloud → remind or share with family.

Honest verdict. The idea is strong and the evidence exists. Three things decide whether it wins: (1) the 500M–700M model must not be the source of medical facts, (2) you must collect your own field evidence early, and (3) the MVP must be narrow enough to demo reliably. Details are in Section 3.

2. The problem, and how we know it is real

The submission's most important question is "What is the specific problem, and how do you know it's real?" This section gives the published evidence found so far. Section 12 gives the field protocol for the evidence only you can collect.

Problem statement (draft). Older people in India often manage several medicines and a steady flow of official papers that they cannot fully read or understand. When help is not at hand, they guess, skip, double up or panic, and scammers use fake notices to exploit exactly this confusion.

Published evidence.

Finding	Source	Note
Pooled across 27 studies, about 49% of older Indian adults have polypharmacy (5–9 medicines), 31% hyperpolypharmacy (10+), and 28% use potentially inappropriate medicines	Frontiers in Pharmacology, 2021 meta-analysis	Strongest source; peer-reviewed
In a six-city study of 600 older adults, 33.7% took five or more medicines daily and 19.7% self-medicated	Cross-sectional study reported by The South First	Check the original paper before quoting
In a rural health-centre study of 310 elderly patients, non-adherence was about 50% and self-medication about 24%	Rohilkhand Medical College Hospital study, 2011	Old and single-site; use as illustration only
Senior citizens filed 103,488 cyber-fraud complaints in 2025, with reported losses of about ₹4,005 crore	MHA reply in Rajya Sabha, 5 Aug 2026, as reported by newsreel.asia	The Rajya Sabha reply is the primary source
In “digital arrest” scams, fraudsters send forged FIRs, bank letters and fake Interpol notices; Pune police recorded 136 such cases and over ₹100 crore lost by seniors in 19 months	PTI via Siasat; Punekar News, 2026	Shows that fake paper is the weapon
Fraud-awareness material and bank alerts mostly default to English or Hindi	MoneyLife, 2026	Supports the local-language angle

What this proves and what it does not. These numbers prove scale and risk. They do not prove the specific gap Saathi fills: people not understanding what is printed on a strip or paper, and being frightened by explanations. That has to come from your own observations (Section 12).

3. Design principles and five hard truths

Principles.

1. Calm first. Never open with a diagnosis, a warning or a disease name.
2. Ask before telling. Find out what the person knows and feels first.
3. Truthful always. Gentle framing is allowed; deception is not. If asked directly, answer directly.
4. Facts from sources, words from the model. Every medical statement must trace to a stored, reviewed field.
5. Never diagnose, never change doses, never say stop or start a medicine.
6. Voice-first, local language, large type, one action per screen.
7. Fail safe: when unsure, say so and send the person to a pharmacist or doctor.
8. Privacy by default: process locally where possible, store nothing unless the person agrees.

Five hard truths (read before building).

1. **A 500M–700M model cannot be your medical memory.** Models this small forget and invent facts, especially brand names and strengths. Google describes its 270M Gemma as built for narrow, task-specific fine-tuning rather than complex conversation, and Qwen3-0.6B sits in a similar size class. Use the model for behaviour (asking the right questions, tone, structured output, rewriting) and a verified database for facts. This also gives you a strong demo moment: the model alone versus the model plus database on the same questions.
2. **“Don’t tell them it is for the heart” needs care.** Your instinct is right: a blunt label can cause panic. But hiding what a medicine is for is paternalistic and can backfire, because the person has a right to know and family or another doctor may tell them anyway. The safer design is gentle disclosure: ask first, explain how to take it first, describe purpose in everyday words, let the person choose how much to hear, and always answer truthfully when asked directly. Section 5 specifies this.

3. **Handwritten prescriptions are the hardest input.** Doctor handwriting defeats most OCR. For the MVP, support printed strips, printed boxes and printed prescriptions. For handwriting, show what was read, ask the person to confirm each line, and send them to the pharmacist if unsure.
4. **Do not identify loose tablets from a photo.** Color and shape matching is unreliable and a wrong answer is costly. Scope is printed packaging only.
5. **Two products in one is a scope risk.** Share one pipeline (photo → understand → explain gently → speak) and build the medicine path deeper. The notice path can launch with three document types.

4. Users and experience

Primary user. A person aged 60+, possibly with low vision, tremor, limited reading ability or unfamiliarity with apps, but comfortable with a phone camera, WhatsApp and voice messages.

Secondary users. A family caregiver or neighbour volunteer who sets things up and receives opt-in updates.

Reviewers (not users). A pharmacist and a doctor who review the medicine rules and the conversation script. Their sign-off is a credibility asset for judges.

Experience rules.

- The home screen has three huge buttons: Medicine, Paper, Ask.
- Voice in and voice out everywhere; every card has Play, Slower and Again buttons.
- The camera screen guides the photo (outline, “hold steady”, auto-capture) and speaks problems aloud (“too dark, please turn on the light”).
- No login for first use; the profile stays on the device.
- “My medicines”: the person’s own strips shown as photos with a simple day schedule (morning, afternoon, night).
- A permanent “Call my family” button and an emergency card (112 and 108).
- Family mode: link by a short code; family can add medicines and see missed-reminder notices only if the older person agrees.
- Delivery: a progressive web app (installable from a link, no store needed) is the fastest for a hackathon. A WhatsApp bot is a strong later channel because older users already use WhatsApp.

5. The Gentle Disclosure Protocol (the heart of the project)

This protocol is the original contribution. Write it as a one-page spec that a doctor reviews, then implement it as a state machine, with the model filling in wording inside fixed rails.

Why it exists. The same fact lands differently depending on what the person already knows and fears. “This is for heart patients”, said to someone who has never been told they have a heart condition, is a shock, not information.

Steps.

Step	What happens	Rule
S0 Capture	Photo of strip, box or prescription. QR/barcode read first, OCR second	Confirm with the picture: “Is this Telma 40?” The person taps Yes or No. Never proceed on an unconfirmed match
S1 Warm open	“Let’s look at this together. How are you feeling today?”	No medical content yet
S2 Ask (max 2 questions)	“What did the doctor tell you this medicine is for?” and “Is anything troubling you these days?” Tap answers: Sugar, BP, Pain, Stomach, Sleep, Don’t remember, Other (voice)	Skippable. Never an interrogation
S3 How to take it	From the label or prescription: when, with food or not, how long	Always first, because it is the most useful and least frightening. Numbers come from the printed text, never from model memory
S4 Purpose in everyday words	If the person named a reason, confirm and match it. If not, offer a choice: “Shall I tell you what this is usually given for, or would you like to ask your doctor?”	If asked directly, answer truthfully. Use the framing table below

Step	What happens	Rule
S5 Companions	Show other medicines often taken with this one; ask “Do you have this one too?” and prompt a photo	Only from the reviewed pairing table
S6 Close	Summary card, reminders, “please do not stop any medicine on your own”, questions to ask the doctor, family share	Always says who to call

Detail layers.

Layer	Content	When shown
1	Name, how to take, expiry	Always, first
2	Purpose in plain words	After S2, or on request
3	Common effects, when to call a doctor	On request; red-flag symptoms stated simply
4	Full technical sheet	Family or pharmacist view

Framing rules.

Instead of	Say
“This is for heart patients.”	“This helps keep your blood pressure steady. Your doctor will know why it suits you.”
“This is a diabetes drug.”	“This helps keep sugar levels steady.”
A full list of side effects up front	“Most people have no trouble. Call your doctor if you notice [one or two red flags].”
“You must not stop.”	“Please do not stop it on your own. If you want to change anything, ask your doctor first.”

Sensitive categories (mental health, HIV, cancer, epilepsy, dementia, addiction treatment and similar): default to neutral wording, do not state a condition unprompted, offer “ask your doctor” as an option, and answer truthfully if the person asks directly. Do not show these on the family view unless the person agrees.

Companion medicines and duplicates.

- Companion table: a short, reviewed list of common pairings (for example, a stomach-protecting tablet often given with certain painkillers, or calcium with vitamin D). These are illustrations only; every row needs pharmacist sign-off and a source.
- Duplicate-ingredient alert: if two strips share an active ingredient (such as two products that both contain paracetamol), flag it kindly: “These two have the same ingredient. Please ask your pharmacist whether you need both.” This is rule-based, cheap and genuinely valuable.
- Interactions: do not claim a full interaction engine. Ship duplicates plus a small reviewed list of well-known risky pairs; otherwise say “ask your pharmacist.”

Example (English with a Hinglish line).

> Saathi: I can see Telma 40. Is that right? [Yes / No] Person: Haan. Saathi: Thank you. How are you feeling today? Did the doctor tell you what this one is for? Person: Doctor ne BP bola tha, shayad. Saathi: Good, that matches. Take one tablet in the morning, at the same time every day. This helps keep your BP steady. Would you like to hear anything else about it?

Hard rules (enforced in code, not left to the model).

1. No diagnosis, no dose changes, no “stop” or “start” instructions.
2. No condition is named unless the person said it or asked.
3. Any statement about purpose, timing or effects must carry a source id from the database.
4. Low match confidence or an unknown medicine: say what was read, ask to confirm, suggest the pharmacist.
5. Danger phrases from the user (chest pain, trouble breathing, fainting, overdose, “took too many”) stop the flow and show emergency steps (112 or 108) and a “call family” button.
6. Every answer ends with an easy way to hear it again and to ask a person.

6. System architecture

Pipeline.

1. Capture: PWA camera with quality checks (blur, glare, darkness).
2. Route: the person taps Medicine or Paper (MVP); auto-classify later.
3. Read: QR/barcode decode first, then OCR; combine the two.
4. Extract: name, strength, form, manufacturer, batch, expiry (medicine) or sender, amount, date, requested action (notice).
5. Match: normalise and fuzzy-match against the local medicine database; show top candidates with confidence.
6. Confirm: the person confirms by tapping the picture.
7. Retrieve: pull the reviewed Medicine Card (facts plus source ids).
8. Converse: the state machine runs the Gentle Disclosure Protocol; the model writes wording from card fields.
9. Validate: the safety layer checks that every claim maps to a card field and no rule is broken.
10. Speak: translate if needed, synthesise speech, play; offer slower and again.
11. Remember: save to “My medicines” only if the person agrees; schedule reminders; family share.

Layers and free options. Benchmark OCR and voice on your own photos before choosing; the options below come from general knowledge and need your own testing.

Layer	Job	Free or open options
Client	Camera, mic, speaker, large-type UI	Progressive web app; browser Web Speech API as fallback; free tiers of Netlify, Vercel or Cloudflare Pages (check current limits)
API	Orchestration	FastAPI (Python); run on a laptop for the demo, or a free Hugging Face Space or Render instance
Codes	QR and barcode reading	ZXing or pyzbar
OCR	Text from strips and papers	PaddleOCR, Tesseract (Hindi plus English), Google ML Kit on device, Surya; test all on 100 real photos

Layer	Job	Free or open options
Matching	Brand → salt → card	SQLite with FTS5 plus RapidFuzz, shipped with the app
Knowledge	Reviewed Medicine Cards	Your own curated JSON or SQLite (Section 8)
Model	Questions, tone, rewriting, extraction	Qwen3-0.6B fine-tuned with LoRA, served as GGUF through llama.cpp or Ollama
Voice	Speech to text, text to speech, translation	Bhashini (free, registration required); AI4Bharat open models as a self-hosted fallback; browser speech
Safety	Rules and validator	Plain Python; no model
Storage	Local profile, optional sync	On-device storage; optional Supabase free tier for family sharing

QR codes help a lot. Since 1 Aug 2023, the top 300 brands under Schedule H2 of the Drugs Rules must carry a barcode or QR code, and the rule is expanding (anti-cancer drugs, vaccines and narcotics by July 2027; antimicrobials by July 2028). Press reports describe the code as carrying brand and generic name, batch, and manufacturing and expiry dates. Caveat: many codes encode a link to a manufacturer page rather than raw data, so treat the QR as one signal and fall back to OCR.

Confidence policy.

- High (exact QR match, or OCR name plus strength matches one entry): show it and ask for confirmation.
- Medium (several candidates): show pictures of the top three and ask.
- Low: say what was read, do not explain the medicine, and suggest the pharmacist.

7. Free data sources

Source	What it gives	Access	Caveats
Jan Aushadhi (PMBI) product list	Official basket of generic medicines with generic name, unit size, MRP and group name; a mirror site reported about 2,100 medicines in Oct 2025	Free; download from the official Jan Aushadhi site (janaushadhi.gov.in)	Covers only the scheme's products, not every brand; use the official site rather than mirrors
Kaggle "Indian Medicine Data" (about 195,600 rows) and "A-Z Medicine Dataset of India" (250K+ rows)	Brand names, salt composition, manufacturer, descriptions, side effects, interactions, prices	Free on Kaggle	Compiled by scraping; check each dataset's licence; prices are dated (Nov 2022); use for brand → salt mapping only and do not ship descriptions without review
IndianDrugMCQA (Eka Care, Hugging Face, MIT licence)	1,512 multiple-choice questions mapping Indian brands to generics	Free	Use as your evaluation set, not as training data
MedlinePlus Connect (NIH / NLM)	Links to patient-friendly drug information; best match with an RxNorm code; English and Spanish	Free, no registration	Content is licensed from ASHP, so link out rather than copy; credit MedlinePlus; no logo; no implied endorsement; mostly US product names

Source	What it gives	Access	Caveats
RxNorm / RxNav and openFDA drug labels	Standard ingredient identifiers and label text	Free APIs (from general knowledge; verify current terms)	US-centric; use at ingredient level only
National List of Essential Medicines and CDSCO notices	Official lists to decide which medicines to curate first	Free PDFs on government sites	For prioritising, not for patient text
Pack QR and barcodes (Schedule H2)	Unique ID, brand and generic name, batch, expiry	Scanned from the pack	Partial coverage; many link to manufacturer pages
Bhashini	Speech to text, text to speech and translation across 22 Indian languages	Free; registration needed	Quotas and uptime unknown; keep a fallback
Official scam guidance	National Cyber Crime Reporting Portal, helpline 1930, RBI and police advisories	Free	Base the scam rules and “what to do” text on these and link to them
Your own field photos	100–200 photos of real strips, boxes, prescriptions, bills and notices, taken with consent and with personal details covered	Free	Your test set, and proof of effort

Do not assume that well-known commercial drug databases (DrugBank, for example) are free for use inside an app; check their licences before relying on them.

8. Building the Medicine Knowledge Base

Depth over breadth. A reviewed set of 150–250 medicines beats 250,000 scraped rows, because every sentence the app speaks must be safe.

1. Choose the 150–250 medicines older people most often take (blood pressure, sugar, cholesterol, thyroid, stomach acid, pain, calcium and vitamin D, blood thinners, common antibiotics), using the essential-medicines list, the Jan Aushadhi list and what you actually see in the field photos.
2. For each medicine, write one Medicine Card (below). Purpose lines are written by you in plain words and reviewed by a pharmacist.
3. Build a brand → ingredient index from the datasets. This is how 250,000 brand names reach your 200 cards.
4. If a brand is not found but the ingredient is known, show ingredient-level information; if neither is known, refuse to explain and point to the pharmacist.
5. Expiry: parse the printed date (formats such as MM/YYYY vary) with rules, then say “this expired in March” or “expires this month.” Never let the model read dates.

Medicine Card (example only; the values are illustrative and not reviewed medical content).

```
id: med_0042
generic: [telmisartan]
strengths: [20 mg, 40 mg, 80 mg]
form: tablet
brands: [Telma, Jan Aushadhi telmisartan]
group: blood-pressure medicine
sensitive: false
purpose_plain: {en: helps keep blood pressure steady, hi: ...}
how_to_take: read from label or prescription only
red_flags: [fainting, swelling of face or lips]
often_with: [med_0107]
duplicate_key: telmisartan
links: {medlineplus: ...}
review: {by: pharmacist name, date: ...}
sources: [src_01, src_02]
```

9. The model: 500M–700M fine-tuning plan

Base model choice.

- Qwen3-0.6B (0.6B parameters, Apache 2.0 licence) is the closest fit to your size range.
- Gemma 3 270M is smaller and designed for narrow fine-tuning, but it is not built for complex conversation and its licence has usage restrictions.
- SmolLM2-360M is another option. Check each licence before release.

Language reality. Tiny models are weak in Hindi and regional languages. The practical design is for the model to work in English (and light Hinglish) while Bhashini translates in and out. Test this in the first days, because it decides the whole voice experience.

What the model does.

1. Slot extraction: turns the person's spoken answer into a category (named a reason, said "don't know", unrelated).
2. Next-move selection inside the protocol: ask, explain layer n, offer a choice, or escalate.
3. Rewriting: turns card fields into one or two short, calm sentences at a fixed reading level.
4. Notice extraction: turns OCR text into structured fields (JSON).

What it must not do. Recall drug facts, doses or interactions, or interpret legal meaning.

Training data.

- 3,000–10,000 short dialogue turns that follow the protocol, generated with a larger model as a teacher using your written spec, then reviewed by you and ideally a pharmacist or doctor (check the teacher model's terms of use).
- 1,000–2,000 synthetic and redacted real notices with labelled fields.
- About 300 adversarial examples: asking for a dose change, asking "am I going to die?", trying to get a diagnosis, asking to hide medicines from family.

Method. LoRA or QLoRA with Hugging Face TRL or Unsloth on a free Colab or Kaggle GPU, then export to GGUF (4-bit) for CPU serving.

Order of work for the hackathon. (1) Build everything with a prompted base model plus rules. (2) Log failures. (3) Fine-tune for behaviour. (4) Show before and after in the demo. If time runs out, you still have a working product.

Evaluation.

Test	How	Why it matters
Brand → generic accuracy	Model alone versus model plus database on IndianDrugMCQA	Your evidence that retrieval is needed

Test	How	Why it matters
OCR field accuracy	Name, strength and expiry on your own photos	Wrong strength (40 versus 4.0) is the dangerous error
Protocol adherence	Share of turns that follow the rules, checked automatically and by hand	Proves the gentle design works
Safety red-team	100 hostile or tricky prompts	Target: zero hard-rule violations
Elder comprehension	Teach-back test with 5 or more people (Section 12)	Real-world proof
Speed	Photo to first spoken word on the demo device	Set a target, then measure it

10. The Notice Decoder module

MVP document types. (1) Electricity, water or gas bill. (2) Bank, insurance or KYC letter, or an SMS screenshot. (3) Government or police-style notice.

Output card (always the same five lines). What is this · What do I need to do · By when · How much · Can I trust it?

How it works. OCR → document type → field extraction (sender, reference number, amount, dates, requested action) using rules first (a date-parsing library for dates) and the model second → each extracted field is highlighted on the photo so the person can verify it → a plain-language summary, read aloud.

Scam check (rule-based first). Flag these signals: threats of arrest or “digital arrest”; being told to stay on a video call; being told to keep it secret; payment to a personal account or UPI ID; requests for OTP, PIN or a remote-access app; deadlines in hours; a claimed authority that does not match the channel; odd spelling or formatting.

Message when flagged. “Police and courts do not arrest people or take money over a video call. Please do not pay or share any OTP. Show this to someone you trust, or call 1930.” Police in several cities have stated that digital arrest has no basis in Indian law.

Rules for this module.

- Never tell the person to call a number printed on the notice; point them to the number on their own bill, card or the official website.
- Do not give legal or tax advice. Say what the paper says and who to ask.
- Show “I may be wrong” wording whenever confidence is low.
- A Share button sends the summary to family.

11. Safety, privacy and legal

Positioning. Saathi is a reading and understanding aid. It does not diagnose or treat. Say this on first run and in the submission. If you ever launch commercially, check whether medical-software rules apply (I am not a lawyer).

Data protection. India’s Digital Personal Data Protection Rules were notified in November 2025 and phase in over about 18 months: consent-manager rules from about November 2026, and the main duties (notice, consent, breach reporting, children’s data) from about mid-May 2027. Health information is sensitive in practice. Design for it now:

- Explicit consent in the person’s own language.
- Collect the minimum; process on the device where possible.
- Do not store photos by default; offer a Delete everything button.
- Family sharing is opt-in and can be switched off at any time.
- No selling of data and no advertising use.

Model safety. Validator layer, hard rules in code, red-team set, logs that contain no personal content, and human review of the script.

Medical review. Ask a pharmacist and a doctor (a family contact, a local pharmacy, a nearby college) to review the companion table, the framing table and the top cards. Record reviewer name and date on each card.

Failure modes.

Failure	Mitigation
Wrong medicine matched	Picture confirmation; confidence tiers; refuse on low confidence
Strength misread (for example 40 versus 4.0)	Strength must match a known strength for that ingredient, else ask

Failure	Mitigation
Translation error in a safety sentence	Pre-translated, reviewed strings for all safety text
Person is in distress	Danger phrases stop the flow and show emergency steps
Over-trust in the app	Repeated, simple reminder: ask your pharmacist or doctor

12. Field research protocol: your “how do you know it is real” evidence

Goal. Talk to 10–15 older adults or their caregivers, 2–3 pharmacists and, if possible, one doctor in your neighbourhood within a few days. Start this now, in parallel with building.

Ethics and safety.

- Go with a family member or teacher, especially when visiting homes.
- Explain what you are building and get a clear yes before asking anything or taking any photo.
- Never photograph someone else’s prescription without permission; cover names and numbers.
- Do not give medical advice; do not ask for health details beyond what they choose to show.
- Keep notes anonymous (R1, R2, R3).

Questions for older adults and caregivers.

1. Which medicines do you take each day, and who told you how to take them?
2. Show me how you decide which tablet is which.
3. When were you last unsure what a medicine was for or when to take it? What did you do?
4. Have you ever missed a dose or taken one twice?
5. When a bill, notice or message arrives that you cannot understand, what do you do and who helps?
6. Have you or someone you know received a suspicious call or message about money, police or the bank?

7. How do you feel when someone tells you the name of an illness or what a medicine is for?

Questions for pharmacists. How often do older customers ask what a tablet is for or when to take it? What are the most common confusions? How often do you see duplicate medicines? How readable are prescriptions?

Observation (with permission). Watch how they sort their medicines. Count how many medicines they take, how many they can name a purpose for, and how many strips are printed in a language or size they cannot read.

Teach-back test. Show one strip and ask: “When do you take this, and what is it for?” Record correct, partly correct or wrong. Repeat after using Saathi. This before-and-after result is your strongest demo evidence.

How to write it up. Use sentences like: “I spoke with N people aged ___ in ___. X of N could not say what at least one of their medicines was for. Y of N had a call or message they were unsure about. Z of N said they would be frightened if told the name of an illness.” Keep exact quotes (with permission) and photos with details covered.

13. Build plan, demo and submission

Phases (no deadline known yet, so order matters more than dates).

Phase	Work
0	Field work and 100 photos (start immediately, runs in parallel)
1	Core pipeline: camera → QR and OCR → match → Medicine Card → spoken output, with a prompted base model
2	Gentle Disclosure state machine, hard rules, Hindi and English voice
3	Notice Decoder (three document types) and scam rules
4	Fine-tune, evaluation, red-team
5	Elder testing, fixes, demo video, README, submission

Cut list (drop in this order if time runs short). Auto-classification of document type → family mode → WhatsApp → languages beyond Hindi and English → fine-tune (keep the prompted version) → third notice type.

Must-have for the demo. Medicine path end to end; the gentle protocol visible on screen; one scam notice caught; one duplicate-ingredient alert; one expiry flag.

Demo video outline (2–6 minutes).

1. 0:00–0:40 The problem, told with your field evidence (quotes, photos).
2. 0:40–1:30 Live: strip → confirm → gentle questions → layered explanation → voice.
3. 1:30–2:30 Companion and duplicate-ingredient alert.
4. 2:30–3:30 Notice: a scam caught.
5. 3:30–4:30 Under the hood: architecture, and model alone versus model plus database.
6. 4:30–5:15 Test with an older person: teach-back results.
7. 5:15–5:45 What you would build next.

Mapping to the submission form.

- Project name: Saathi (or Dawa-Saathi if you want the medicine focus visible).
- Elevator pitch: Section 1.

- Problem and evidence: Sections 2 and 12 (lead with your own observations, then the published numbers).
- What it does and how: Sections 5 and 6.
- Built with: PWA, FastAPI, OCR engine, SQLite, Qwen3-0.6B with LoRA, Bhashini, plus whatever you finally choose.
- Challenges: Section 3 (small model, handwriting, OCR errors, language, safe wording).
- Next steps: WhatsApp channel, more languages, pharmacist review network, on-device model, caregiver features.
- Public repository: README with architecture, setup, synthetic sample data and the scripts that build the database. Do not commit real people's photos or API keys.

14. Open questions for you

1. What is the submission deadline, and how many days do you have?
2. Which languages matter most for the people you can reach (Hindi, English, or a regional language)?
3. Can you reach 10 or more older adults this week, and a pharmacist?
4. Which devices will you demo on (Android phone, laptop)?

15. Sources and notes

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- Schedule H2 QR codes: downtoearth.org.in/health/how-qr-codes-on-antibiotics-can-strengthen-indias-fight-against-amr; medianama.com/2025/08/223-govt-qr-code-top-300-pharma-brands/; CDSCO guidance on spurious drugs (cdsco.gov.in)
- DPDP Rules 2025 timeline: hoganlovells.com/en/publications/indias-digital-personal-data-protection-act-2023-brought-into-force-; azbpartners.com/?p=87799
- Small models: huggingface.co/Qwen/Qwen3-0.6B; The Register on Gemma 3 270M: theregister.com/2025/08/15/little_llm_on_the_ram/
- Bhashini: bhashini.gov.in (free registration for developers)

Not verified in this research (from general knowledge; test before relying): OCR library accuracy on Indian packaging, RxNav and openFDA terms, free-tier limits of hosting services, and the status of individual Bhashini services. Statistics marked as secondary should be checked against the original paper or government reply before you quote them.

16. Addendum: handwritten prescriptions (expands hard truth 3)

The honest target. Not “read every prescription”, but “narrow every line to a few likely medicines, confirm with a second source, and fail safely.” Even trained pharmacists sometimes phone the clinic; the app should behave the same way.

What the research says.

- MIRAGE (arXiv 2410.09729): fine-tuned multimodal models (LLaVA 1.6, Idefics2) reached about 82% accuracy on medication names and dosages across 743,118 simulated Indian prescription records from 1,133 doctors. The handwriting was collected on a tablet and is unusually clean, so real photos will score lower. Earlier systems reported roughly 64–74%.
- The same work found that giving the model the doctor’s specialty and commonly prescribed medicines improved accuracy.
- An April 2026 benchmark on RxHandBD (5,578 cropped handwritten words) found newer compact OCR engines (PP-OCRv5, GLM-OCR) clearly better than Tesseract and EasyOCR, but still well short of what unsupervised clinical use needs. It tested cropped words, not whole pages, so page-level reading is harder.

- Implication: expect about one line in five to be wrong or unreadable at best. A wrong answer shown with confidence is the dangerous case, so the design is built around confirmation, not around accuracy alone.

The Prescription Reconciliation pipeline.

1. Layout: separate the printed header (clinic, doctor, specialty, date) from the handwritten body, then split the body into lines. One line is normally one medicine entry.
2. Reading: normal OCR for printed parts; a handwriting reader for each line that returns the top five guesses with scores, never a single answer.
3. Dose grammar: parse form (Tab, Cap, Syp, Inj, Drops), frequency (1-0-1, OD, BD, TDS, HS, SOS), timing (AC, PC) and duration (× 5 days) with a small rule set. These tokens are short and repetitive, so they are often readable when the name is not, and they filter candidates (a medicine with no syrup form cannot be a Syp line).
4. Lexicon snapping: match the guesses against your brand and generic list using edit distance and phonetic matching, boosted by the doctor's specialty and by medicines commonly prescribed together.
5. **Strip reconciliation (the key step).** Ask the person to photograph the strips or boxes they actually bought. Printed strips are easy to read. A strip whose ingredient appears in a line's top candidates confirms that line. A line with no matching strip gets "I could not match this one; please ask your pharmacist." A strip that matches no line gets "this is not on the list I could read; please check with your pharmacist." This also catches missing and extra medicines.
6. Human confirmation: large cards saying "Is it one of these?" with the candidate names and typical strip pictures. A pharmacist card shows the cropped handwritten line beside the candidate names so a pharmacist can confirm in seconds. A "Call the clinic" button appears when a number is printed in the header.
7. Safe failure: a line that is not confirmed never gets purpose, timing or warnings, only "I am not sure about this one."

Look-alike, sound-alike guard. From your lexicon, precompute pairs of names that are close in spelling but have different ingredients. If a line's top two candidates form such a pair, require strip or pharmacist confirmation, whatever the confidence score says.

Which model reads the handwriting. The handwriting reader is a vision model, separate from your 0.6B language model, which only handles the conversation and the parsing of the reader's output. Options to test on your own photos: a fine-tuned open multimodal

model as in MIRAGE; compact OCR for line crops; a hosted multimodal model API as a reader or fallback if its terms and free tier allow it (verify; the image leaves the device, so ask for consent first).

Data. Public sets (Doctor's Handwritten Prescription BD with 4,680 word crops and 78 medicine names; RxHandBD; the 100 labelled records released with MIRAGE) come mostly from other settings or are simulated, so treat them as warm-up. Build your own set: 30–50 real prescriptions from your area, collected with consent and with personal details covered, transcribed by a pharmacist. That set is both your test data and your field evidence.

Success metrics.

- Top-3 contains the true medicine for X% of lines (set X after the first test).
- Zero lines shown as confirmed that are wrong.
- Share of lines confirmed through a strip.
- Time a pharmacist needs to confirm a line using the share card.

Prevention nudge. Include a polite card the person can show at the clinic: "Could you please write the medicine names in capital letters, or print the prescription?" Printed or digital prescriptions are always the best input.

Demo idea. Show a messy prescription: three lines confirmed by strips, one flagged as unreadable, then the pharmacist card. This turns the weakness into your best safety story.

17. Addendum: the "Save money" module

Where it sits. A separate tab, available only after a medicine is confirmed and explained. Never inside the gentle conversation, so the app does not feel like a sales funnel.

What it shows, in order.

1. **Same-ingredient options**, only when active ingredient, strength and form match exactly. Wording: "These contain the same ingredient. You can ask your doctor or pharmacist whether one of them suits you." Never "switch." Combination products often have no exact match, and the app should say so.
2. **Jan Aushadhi**: if a matching generic exists, show its MRP next to the brand MRP and the nearest Jan Aushadhi Kendra. The Jan Aushadhi Sugam app and an all-India Kendra list are published; check the official site for the latest file.

3. **Overcharge check**: NPPA sets ceiling prices for scheduled (essential) medicines, and their MRP must not exceed the ceiling. If the strip or bill is higher, show how to complain on Pharma Jan Samadhan. Pharma Sahi Daam also compares brands with the same ingredient. I did not find a public API for it, so verify with NPPA or NIC; otherwise link out or use a downloaded price list.
4. **Licensed online pharmacies**: link out only, with no ordering or payment inside Saathi.

Guardrails.

- **Legal.** Schedule H, H1 and X medicines may be sold only on a valid prescription from a registered practitioner. The sources I found say specific e-pharmacy rules were still in draft (the 2018 draft was not notified at the time of those sources), and a Delhi High Court interim order in December 2018 directed the Centre to ensure that unlicensed online sale is prohibited. The position may have changed, so check it and ask a legal mentor. Design so the module can never help someone buy a prescription-only medicine without a prescription: offer only links to sellers that require a prescription upload.
- **Define "verified" in writing.** For example: displays a state drug licence number, has a registered pharmacist, asks for a prescription for scheduled drugs, uses a secure site, and is a known company. Keep a short manual whitelist of 3–5 sellers, show the date last checked, and re-check regularly.
- **Scam safety.** Fake pharmacy sites and payment links are a scam route for older people. Run the Notice Decoder's scam checks on any pharmacy link or message, and never send payment links from the app.
- **No scraping** of retailer prices (terms of use, breakage, wrong numbers). Prefer official lists, show "price as of (date)", and say that prices change.
- **No affiliate commissions or sponsored ranking.** If you ever earn revenue from this module, disclose it.
- **Offline first for older users.** A "Show this list at the shop" card for the local chemist or the nearest Jan Aushadhi Kendra; many older people prefer buying in person. Government material says Jan Aushadhi medicines are procured from WHO-GMP certified suppliers and quality tested before dispatch.

Evidence to collect (Section 12 add-on). Ask older adults and caregivers about monthly medicine spending, whether they have heard of Jan Aushadhi or Pharma Sahi Daam, whether they ever bought medicines online, and what worries them about it.

Build order. MVP: Jan Aushadhi price and nearest Kendra, plus the overcharge-check link. Then same-ingredient options. Online pharmacy links last. In the cut list, put this module after the Notice Decoder.

Extra sources. arxiv.org/abs/2410.09729 (MIRAGE); docs.lancedb.com/datasets/handwriting-ocr (Doctor's Handwritten Prescription BD); thearabianpost.com/modern-ocr-narrows-gap-on-doctors-handwriting/ (April 2026 benchmark report); ijme.in/wp-content/uploads/2025/05/Emerging-epharmacy_288.pdf and blog.ipleaders.in/need-regulation-e-pharmacies-india/ (e-pharmacy law); policycircle.org/industry/e-pharmacies-need-to-remove-policy-vacuum/amp/ (Delhi High Court order); pharma-dept.gov.in (NPPA price fixation and Pharma Sahi Daam); newsbytesapp.com (July 2026 explainer on Pharma Sahi Daam); manoramayearbook.in and pmbi.co.in (Jan Aushadhi quality and product basket).